

INTRODUCTION

CREATED BY: DIONTE CAPLETON



- This dashboard examines workforce structure and financial trends across departments and countries.
- The dataset includes 107 employees, 27 departments, and 23 locations.
- Objective: uncover HR risks and cost drivers to inform future hiring and budget planning
- Tools: SQL, Power BI, Python

RESEARCH QUESTIONS



- What factors influence employee retention in the company?
- How do hiring trends affect workforce demographics?
- In what ways are salary trends impacting financial outcomes?

DATA MODELING

DATA SOURCE:



Fictional HR dataset,
107 employees, 7
interconnected
tables.

TOOLS USED:

SQL



Python



Power BI



PROCESS:

- Structured and interconnected tables using SQL.
- Python (Pandas) for data cleaning and visualization.

Structured and interconnected tables using SQL.

Python (Pandas) for data cleaning and visualization.
Analyzed employee counts, hiring trends, and salary spending

HR WORKFORCE OVERVIEW

TOTAL EMPLOYEES

107

AVG SALARY

\$6.46K

TOTAL DEPARTMENTS

27

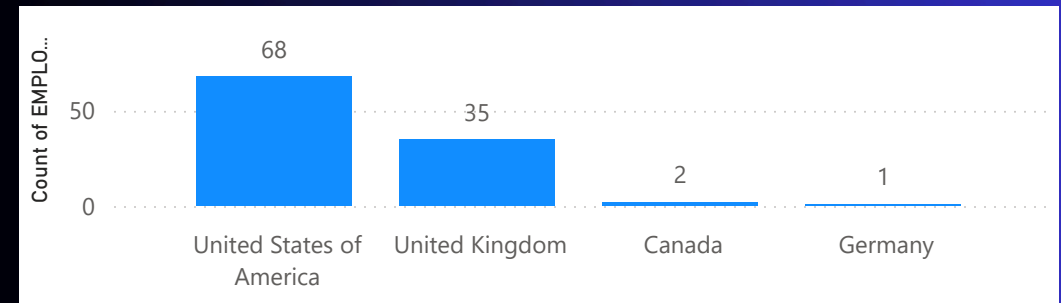
TOTAL LOCATIONS

23

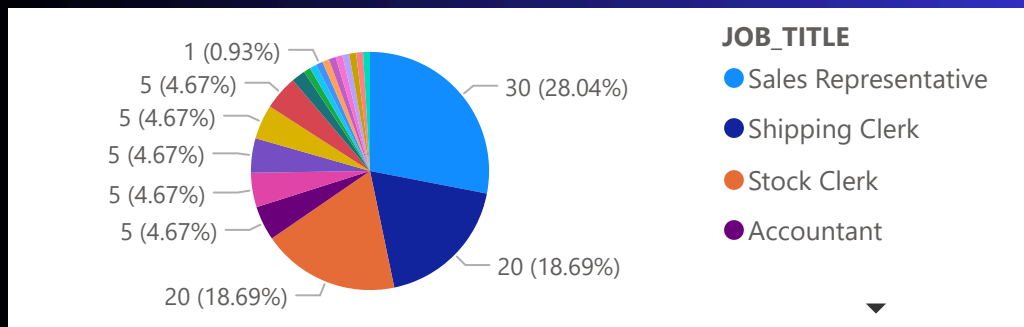
EMPLOYEES BY DEPARTMENT



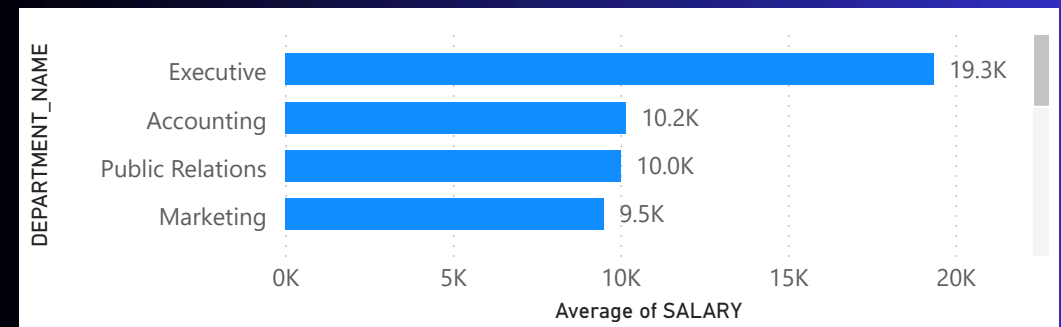
EMPLOYEES BY COUNTRY



EMPLOYEES BY JOB TITLE



AVE SALARY BY DEPARTMENT



HIRING TREND ANALYSIS

TOTAL HIRES

107

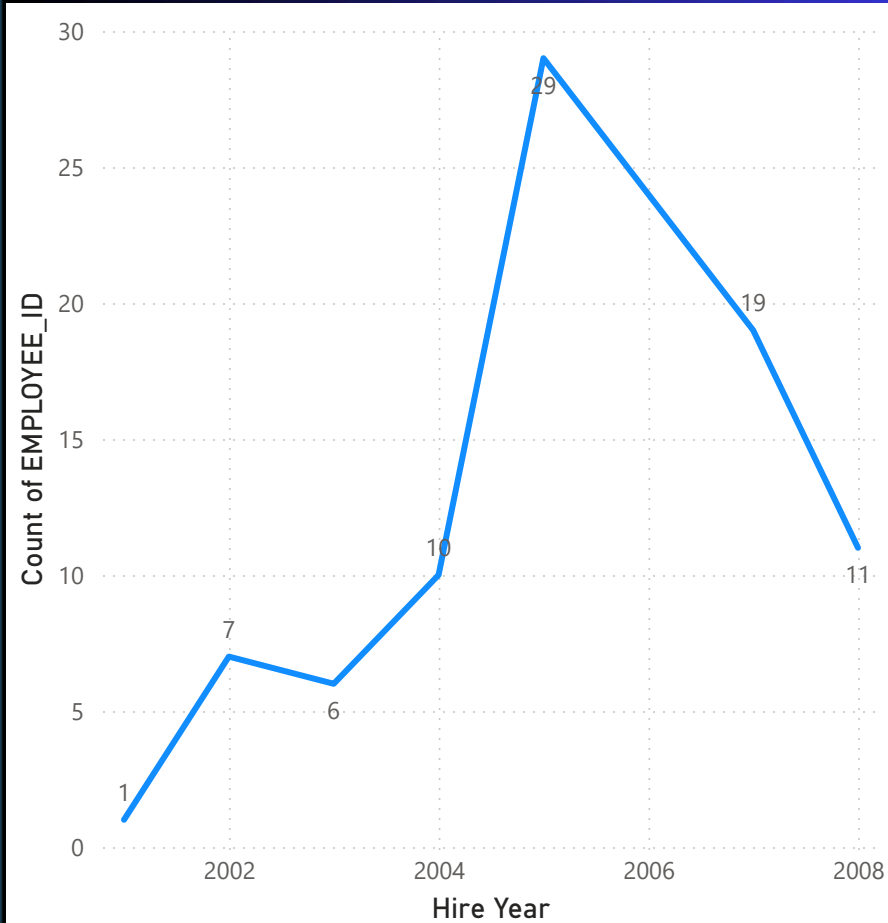
PEAK HIRING
YEAR

2005

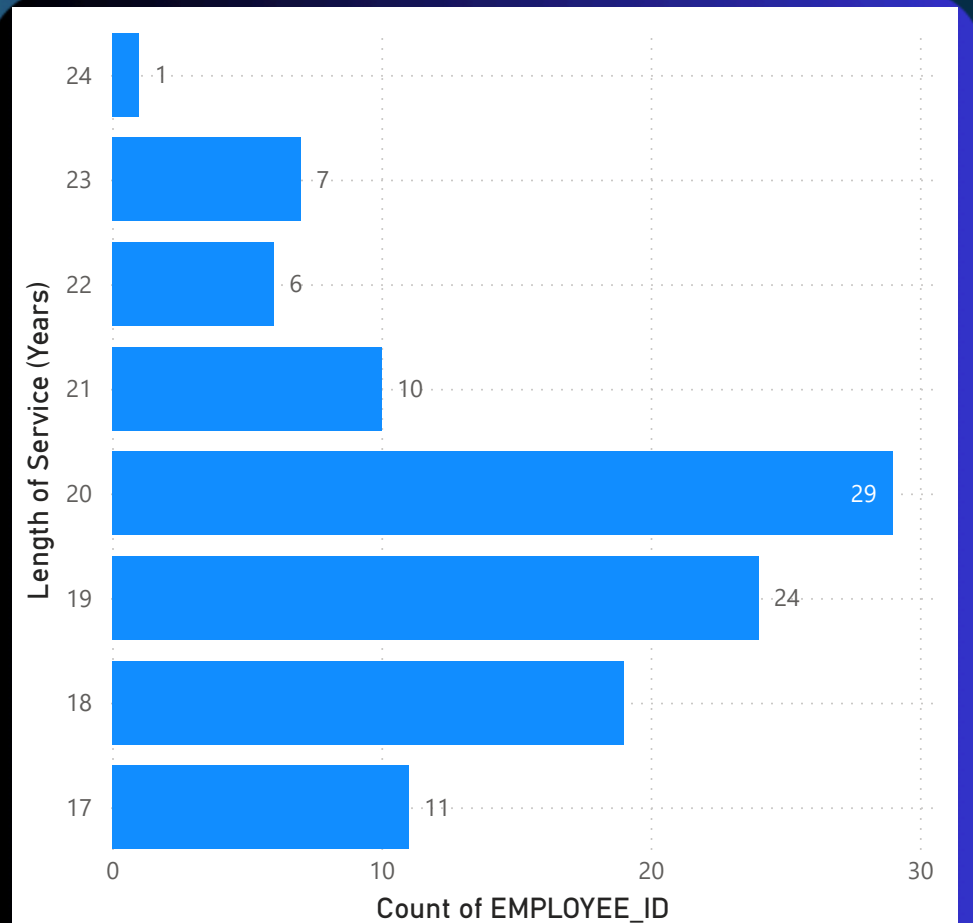
AVG HIRE PER
YEAR

13

NEW HIRES OVER TIME



EMPLOYEES BY LENGTH OF SERVICE



WORKFORCE FINANCIAL OVERVIEW

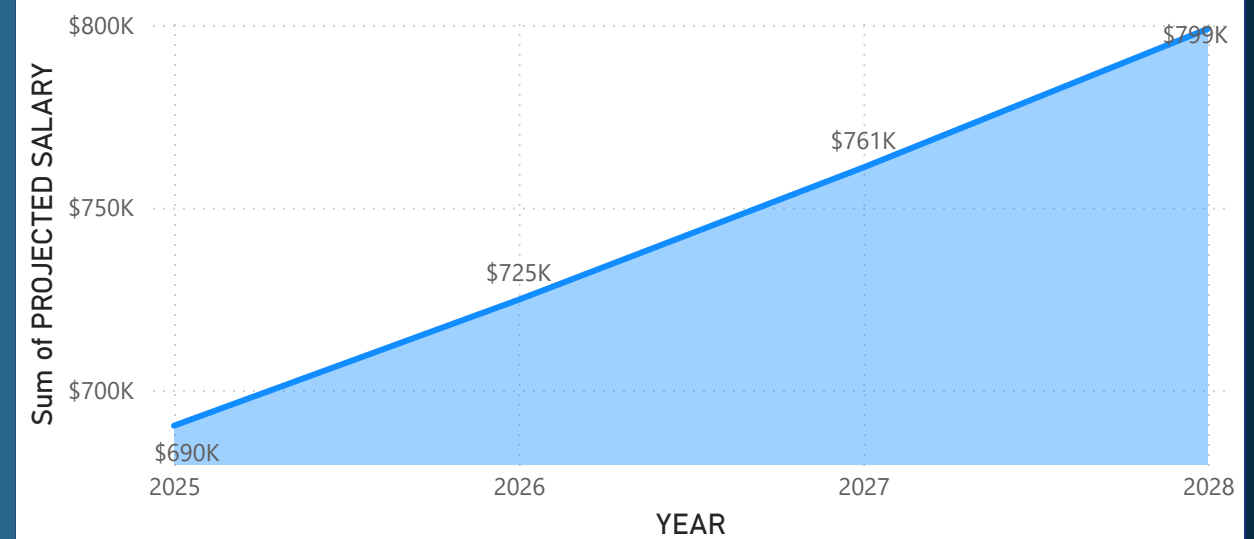
TOTAL ESTIMATED SALARY SPEND

\$691K
ANNUALLY

882.44K

Projected Salary Spend (5 Years)

PROJECTED SALARY COST OVER TIME



**WITHOUT YOUNGER HIRES, SALARY COSTS EXPECTED TO RISE TO
20% BY 2028**

BUSINESS CHALLENGE ANALYSIS

THE CHALLENGE

Given the aging workforce and rising salary costs, how can the company minimize financial risk while maintaining organizational knowledge?

THE SOLUTION

- 100% of employees have over 17 years of service
- Salary costs projected to rise 20% by 2028
- No junior hiring pipeline currently exists

SOLUTION:

- Begin new hiring immediately, targeting 30% workforce renewal over next 3 years
- Balance junior and senior hiring to manage salary costs
- Set up training and shadowing

CONCLUSION

- Long tenures indicate a loyal but aging workforce with a high attrition risk in the future.
- An influx of new hires is needed to rejuvenate the staff and balance senior expertise with fresh skills.
- Projected salary costs are set to reach \$882K yearly, increasing financial pressure.
- Data analysis underscores the need for strategic hiring to mitigate costs and sustain workforce expertise.